

DIRECTIONS for questions 1 to 4: The passage given below is followed by a set of **four** questions. Choose the best answer to each question.

The Viking warriors swept down from the northern most part of the globe like ravenous wolves, spreading terror and bloodshed wherever they went. From their Scandinavian homelands, they ranged as far south as Spain and as far east as Russia. Throughout Europe, people trembled in fear at the very mention of their name.

The Viking raids began in the ninth century, and the destruction they wrought greatly contributed to the turmoil experienced by western Europe during the Dark Ages. The French region of Normandy was conquered in the tenth century by Viking warriors whose descendants achieved important military victories in the 1000s, including the Norman Conquest of England in 1066 and the liberation of Sicily from the Muslims in 1091. Around the turn of the millennium, Viking sailors from Iceland discovered North America and established settlements on its coast – the first Europeans to do so.

Clearly, the exploits of the Vikings had positive and negative effects on Europe, and their accomplishments were essential in shaping the course of European history. Yet as the year 1000 dawned, the Viking age was already nearing its end. Across Scandinavia, the pagan worshipers of Odin and Thor began to convert to Christianity; as good Christians, they became reticent to attack the churches and monasteries that they had once plundered with abandon. The growth of strong centralized monarchies in Norway, Denmark, and Sweden also ultimately resulted in taming the Viking spirit. As Scandinavia became increasingly civilized, its kings discouraged the activities of roaming warrior bands, and its social environment gave rise to a more humdrum life. By the middle of the eleventh century, the glory days of the Viking raiders had faded into twilight, never to return. ...

In the year 1000, Vikings attempted to settle on the North American coast. The earliest written source having this information is not the known saga accounts but *The History of the Archbishops of Hamburg*, completed by Adam of Bremen in about 1075. Two saga accounts, arising out of a different tradition, support this information. The *Greenlanders' Saga*, the earlier of the two was committed to writing in the twelfth century and has about it a primitive crudeness which, while not particularly attractive literally, adds to its historical credibility. The great anthology of Icelandic material, the Flatey Book, compiled towards the end of the fourteenth century in northern Iceland, contains the earliest extant text of this saga. The *Eric Saga* has a more polished appearance and dates, in its earliest written form, from the mid-thirteenth century, but exists only in two later medieval versions. . . . The *Greenlanders' Saga* and the *Eric Saga* tell essentially the same story, yet in some places they complement and in other places they contradict one another. The *Greenlanders' Saga* is more reliable and its text and story more faithful to an oral original.

Q1. Which of the following can be understood about the Viking warriors?

- ☐ a) Their exploits had greater positive effects on Europe than negative ones.
- ☐ b) Their raids began in the 1000s and they became more unruly and barbaric as the glue of custom and tradition in the native homelands was lost.
- ☒ c) **They belonged to the Scandinavian homelands and were the first Europeans to settle in North America.** ✓ Your answer is correct
- ☐ d) They were not so feared by the people of Europe as the bond that held them together was broken and they became isolated.

Q2. Which of the following statements correspond to what is stated in the passage regarding the end of the Viking age?

- ☒ a) **As good Christians, the converted Vikings eschewed violence.** ✖ **Your answer is incorrect**
- ☐ b) The end of the 10th century was known to be the end of the glorious days of the Vikings.
- ☐ c) **The kings of Spain and Iceland discouraged the activities of the Vikings.**
- ☐ d) The growth of monarchies in certain countries was partly responsible for the end of the Viking Age.

Q3. Which of the following can be inferred from the phrase “its social environment gave rise to a more humdrum life” (para 3)?

- ☐ a) **The future of a race is determined by its historical past and present social milieu.**
- ☒ b) **As Scandinavians became more civilized, the environment became more peaceful.** ✓ **Your answer is correct**
- ☐ c) **The environment of Scandinavia was better than that of Spain in terms of daily living.**
- ☐ d) **The environment of Scandinavia became more peaceful like never before.**

Q4. Which of the following can be inferred about the sagas discussed in the passage?

- ☐ a) **The *Eric Saga* provides cruder details than the *Greenlanders' Saga*.**
- ☒ b) **Flatey's Saga, *Greenlander's Saga*, *Eric's Saga* and Hamburg's Saga are the four saga accounts of the attempted Viking settlement in America as mentioned in the passage.**
- ☐ c) **The *Eric Saga* was more literally attractive and more reliable than the *Greenlanders' Saga*.**
- ☐ d) The *Eric Saga* repeats some of the details of the earlier *Greenlanders' Saga* but also adds more details and changes others.

In his seminal 1950 paper, Computing Machinery and Intelligence, Alan Turing began by considering the question: “Can machines think?” He then went on to suggest that this question is difficult to answer directly, and he turned it around into a research experiment that he called ‘The Imitation Game’, but which subsequent generations simply know as the Turing test for Artificial Intelligence.

The basis of the test is whether a person sitting at a terminal communicating with another person in one room and a computer in another room can tell, in a reasonable length of time, which of their correspondents is the computer and which is the human. If most people sitting at the terminal cannot get this right most of the time, then the computer is judged to have passed the Turing test.

Turing wrote this paper only two years after the world’s first operational electronic stored-program computer ran its first program in Manchester, on June 21, 1948. Indeed, it was this machine that brought Turing to Manchester, and it was during his time here that he wrote this paper. Turing thought that all that a machine would require to pass his test was more memory - the 1948 Manchester ‘Baby’ was quite powerful enough already. He estimated that a gigabyte (a thousand million bytes) of memory should suffice, and this should be achievable by the end of the twentieth century.

By the beginning of the 21st century computers did, indeed, typically have a gigabyte of memory, and they were a million times faster than the ‘Baby’, but still they could not pass his test. Even today, with still far more computing power and memory, no machine has convincingly passed the test. This would have surprised Turing had he lived to see it.

Although research into artificial intelligence (AI) has delivered in many areas of life, it has failed to deliver as expected, particularly by many imaginative science fiction writers, in the area known as Artificial General Intelligence. This is the idea that a suitably programmed machine might display aspects of intelligence that we normally associate only with humans. My take on the failure to deliver this form of AI is that we have never actually worked out what natural intelligence is, so we don’t know what it is that we are trying to imitate in our machines.

If we do build computers capable of mimicking human intelligence they will, at least initially, be the size of aircraft hangars and consume a million times more power than the rather neat version that you carry around in your head. So they would be a very inefficient substitute for the real thing.

Why then, you might reasonably ask, bother trying to build computer models of the brain at all? There are three answers to this question. Firstly, this is a very effective way to advance the science and the quest to understand our own brains and minds remains as one of the great frontiers of science. As the late, great scientist Richard Feynman once said: “What I cannot create, I do not understand.” Secondly, a computer model of the brain would be very useful for understanding diseases of the brain, which is vital for developing new treatments.

Thirdly, understanding the brain is likely to lead to insights that can be used to build better and more efficient computer systems.

Q5. According to the author, what is the reason that we are not able to create an effective Artificial General Intelligence?

- ☐ a) **We have not been able to completely understand Artificial Intelligence which is requisite for creating Artificial General Intelligence.**
- ☐ b) **We do not understand how our own brains work and without this understanding, it is not possible to create an effective Artificial General Intelligence.**
- ☐ c) **We first need to understand how natural intelligence helps computers pass the Turing Test.**
- ☐ d) We have not been able to understand how the Turing test works which is critical for creating Artificial General Intelligence.

Q6. It can be inferred from the passage that Turing made a mistake in assuming that

- ☐ a) **computers will have a gigabyte of memory by the twentieth century.**
- ☐ b) **we will be able to understand how our brains work by the twentieth century.**
- ☐ c) **a large amount of memory is sufficient to pass the Turing test.**
- ☐ d) Artificial Intelligence can be achieved by the twentieth century.

Q7. According to the passage, which of the following is a feature of Artificial General Intelligence but not of Artificial Intelligence?

- ☐ a) **Making complex calculations faster than the average human.**
- ☐ b) **Recognizing the meaning of the words spoken.**
- ☐ c) **Exhibiting intelligence normally associated with humans.**
- ☐ d) Understanding the emotional state of a person from his/her voice.

Q8. Which of the following can be definitely inferred from the passage about Manchester 'Baby'?

- ☐ a) **Manchester 'Baby' was the most powerful computer of its time.**
- ☐ b) **Manchester 'Baby' had less than a gigabyte of memory.**
- ☐ c) **Manchester 'Baby' was not able to pass the Turing test.**
- ☐ d) Manchester 'Baby' had a better chance of passing the Turing test than the machines of the 21st century.

DIRECTIONS *for questions 9 to 12:* The passage given below is followed by a set of **four** questions. Choose the best answer to each question.

Hackers, as a rule, love wordplay and are very conscious and inventive in their use of language. These traits seem to be common in young children, but the conformity-enforcing machine we are pleased to call an educational system bludgeons them out of most of us before adolescence. Thus, linguistic invention in most subcultures of the modern West is a halting and largely unconscious process. Hackers, by contrast, regard slang formation and use as a game to be played for conscious pleasure. Their inventions thus display an almost unique combination of the neotenous enjoyment of language-play with the discrimination of educated and powerful intelligence. Further, the electronic media which knit them together are fluid, 'hot' connections, well adapted to both the dissemination of new slang and the ruthless culling of weak and superannuated specimens. The results of this process give us perhaps a uniquely intense and accelerated view of linguistic evolution in action.

Hacker slang also challenges some common linguistic and anthropological assumptions. For example, in the early 1990s it became fashionable to speak of 'low-context' versus 'high-context' communication, and to classify cultures by the preferred context level of their languages and art forms. It is usually claimed that low-context communication (characterized by precision, clarity, and completeness of self-contained utterances) is typical in cultures which value logic, objectivity, individualism, and competition; by contrast, high-context communication (elliptical, emotive, nuance-filled, multi-modal, heavily coded) is associated with cultures which value subjectivity, consensus, cooperation, and tradition. What then are we to make of hackerdom, which is themed around extremely low-context interaction with computers and exhibits primarily "low-context" values, but cultivates an almost absurdly high-context slang style?

The intensity and consciousness of hackish invention make a compilation of hacker slang a particularly effective window into the surrounding culture — and, in fact, this one is the latest version of an evolving compilation called the 'Jargon File', maintained by hackers themselves since the early 1970s. This one (like its ancestors) is primarily a lexicon, but also includes topic entries which collect background information on hacker culture that would be awkward to try to subsume under individual slang definitions.

Though the format is that of a reference volume, it is intended that the material be enjoyable to browse. Even a complete outsider should find at least a chuckle on nearly every page, and much that is amusingly thought-provoking. But it is also true that hackers use humorous wordplay to make strong, sometimes combative statements about what they feel. Some of these entries reflect the views of opposing sides in disputes that have been genuinely passionate; this is deliberate. We have not tried to moderate or pretty up these disputes; rather we have attempted to ensure that *everyone's* sacred cows get gored, impartially.

Q9. When compared to other Western subcultures, the hacker subculture

- ☐ a) **was slow in disseminating new slang words but very quick in culling out weak and superannuated specimens.**
- ☐ b) **found it pleasurable to constantly invent new words and add them to their slang vocabulary.**
- ☐ c) **employed slang to suggest the breaking of the rules of social etiquette and the class system.**
- ☐ d) invented language through a halting and largely unconscious process.

Q10. The last sentence of the passage “We have attempted to ensure that *everyone’s* sacred cows get gored, impartially” suggests that

- ☐ a) the Jargon File keepers possessed an incredible ability to moderate disputes that cropped up from time to time.
- ☐ b) the use of amusingly thought-provoking and humourous wordplay is the name of the Jargon File keepers’ game.
- ☐ c) the Jargon File contains slang entries that will offend the sensibilities of almost all those whose jargon is being discussed.
- ☐ d) the Jargon File usually contains slang that is nuance-filled, multi-modal and heavily coded but strong, combative statements made by hackers about how they feel also require an intelligent play on words.

Q11. According to the passage, hackerdom is characterized by

- ☐ a) **Low-context communication, low-context values and high-context slang style.**
- ☐ b) **High-context communication, high-context values and high-context slang style.**
- ☐ c) **Low-context communication, high-context values and high-context slang style.**
- ☐ d) Low-context communication, high-context values and low-context slang style.

Q12. From the passage, all of the following can be considered as true about “hacker slang” EXCEPT:

- ☐ a) **Hacker slang evolves continuously, and the compilation of its evolving forms is called the Jargon File.**
- ☐ b) **The formation of hacker slang displays a combination of the enjoyment of language-play with the discrimination of educated and powerful intelligence.**
- ☐ c) **Hacker slang allows the hacker culture to be classified by the preferred context level based on common linguistic and anthropological assumptions.**
- ☐ d) Hacker slang uses humorous wordplay in strong statements making it amusingly thought-provoking.

DIRECTIONS *for questions 13 to 16:* The passage given below is followed by a set of **four** questions. Choose the best answer to each question.

Thrown on to a helpless defensive by changes even in a slowly moving world of agriculture, medieval social theory found itself swept off its feet after the middle of the century by the swift rise of a commercial civilization, in which all traditional landmarks seemed to be submerged. The issue over which the struggle between the new economic movements of the age and the scheme of economic ethics expounded by churchmen was most definitely joined, was not that of wages, but that of credit, money-lending and prices. The center of the controversy was the problem which contemporaries described by the word 'usury'.

In spite of property, Tudor England was still a Distributive State. It was a community in which the ownership of land, and of the simple tools used in most industries, was not the badge of a class, but the attribute of a society, and in which the typical worker was a peasant farmer, a tradesman, or a small master. In this world of small property-owners, the wage-earners were a minority and being, normally themselves the sons of peasants, with the prospect of stepping into a holding of their own, were often in a strong position vis-a-vis their employers.

The special economic malaise of an age is naturally the obverse of its special qualities. The grievance, which supplied fuel to social agitation, sprang not from the exploitation of a wage-earning proletariat by its employers, but from the relation of the producer to his landlord, the dealer with whom he bought and sold, and the local capitalist to whom he ran into debt. The farmer must borrow money when the season is bad. The craftsman must buy raw materials on credit and get advances before his wares are sold. ... Almost everyone has need of the money-lender who is a monopolist – 'a money-master', a corn monger, a rich priest.

It is not surprising that there should have been a popular outcry against extortion. The doctrine as to the ethics of economic conduct, which had been formulated by medieval Popes was rehearsed by the English divines of the sixteenth century, not merely as the conventional tribute paid by a formal piety to the wisdom of the past, but because the swift changes of the period in commerce and agriculture had, not softened, but accentuated, the problems of conduct for which it had been designed. Nor was it only against the particular case of the covetous moneylender that the preacher and the moralist directed their arrows. The essence of the medieval scheme of economic ethics had been its insistence on equity in bargaining – a contract is fair, St. Thomas had said, when both parties gain from it equally. The prohibition of 'usury' had been the kernel of its doctrines, not because the gains of the money-lender were the only species, but because, in the economic conditions of the age, they were the most conspicuous species, of extortion.

In reality, alike in the Middle Ages, the word 'usury' had not the specialized sense that it carries today.... If the common man was asked why usury was wrong, he would have quoted from scripture. If he was asked for a definition of usury, he would have repeated the words of a member of parliament: "It standeth doubtful what usury is; we have no true definition of it."

Q13. According to the passage, reforms in medieval ethical doctrine were required because

- ☐ a) **the sweeping changes could not do away with the focus that affected ethical conduct.**
- ☐ b) **there was a great progress in the field of commerce and a conventional tribute needed to be paid to the wisdom of the past.**
- ☐ c) **the new changes had helped in solving the problems of conduct.**
- ☐ d) the vast changing scenario had aggravated the problems of conduct.

Q14. From the sentence, “The special economic malaise of an age is naturally the obverse of its special qualities” (para 3), we infer that the author is referring to that aspect of society where

- ☐ a) the serious economic problems of an age are the reverse of its noteworthy traits.
- ☐ b) independence in vocation and dependence on external assistance go hand in hand.
- ☐ c) the remarkable economic feature of the society generally has an opposing trend.
- ☐ d) the distinctive aspect of a community is offset by its unusual virtuosity.

Q15. According to the passage, 'Tudor England' was called a Distributive State because

- ☐ a) the capitalists were the proprietors of the industries and they played a role of rich landlords in medieval England.
- ☐ b) most working people owned material resources.
- ☐ c) an average worker was a peasant, a tradesman or a craftsman and land was freely distributed to them.
- ☐ d) the wage-earners were a minority and normally themselves the sons of peasants

Q16. In the Middle Ages, the common man could not explain why 'usury' was wrong because

- ☐ a) **The word 'usury' had different connotations as it was a stretchable term.**
- ☐ b) **The scriptures did not guide him adequately and hence he remained confused.**
- ☐ c) **St. Thomas gave varied interpretations of 'usury' in his doctrines and St. Raymond's manual provided different definitions of 'usury'.**
- ☐ d) A member of parliament claimed that there was no true definition of 'usury'.

Q17. DIRECTIONS *for question 17:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.
Sentence: Only by making certain approximations can solutions be found.

Paragraph: The idea of gravitational waves emerged from the general theory of relativity, Albert Einstein's fundamental exposition of gravity, unveiled almost exactly 100 years ago. Mass, Einstein realized, deforms the space and time around itself. ____ (1) ____ Gravity is the effect of this, the behaviour of objects dutifully moving along the curves of mass-warped space-time. It is a simple idea, but the equations that give it mathematical heft are damnably hard to solve. ____ (2) ____ And one such approximation led Einstein to an odd prediction: any accelerating mass should make ripples in spacetime. However, Einstein was not happy with this idea. He would, himself, oscillate like a wave on the topic – rescinding and remaking his case, arguing for such waves and then, after redoing the sums, against them. ____ (3) ____ But, while he and others stretched and squeezed the maths, experimentalists set about trying to catch the putative waves in the act of stretching and squeezing matter. ____ (4) ____ But the waves themselves proved elusive until the construction of the Advanced Laser Interferometer Gravitational-wave Observatory (LIGO).

- ☐ a) **Option 1**
- ☐ b) **Option 2**
- ☐ c) **Option 3**
- ☐ d) Option 4

Q18. DIRECTIONS *for questions 18 and 19:* The passage given below is followed by four summaries. Choose the option that best captures the essence of the para.

The past is already in debt to the mismanaged present. And besides, contrary to what you may have heard or learned, the past is not done and it is not over, it's still in process, which is another way of saying that when it's critiqued, analysed, it yields new information about itself. The past is already changing as it is being re-examined, as it is being listened to for deeper resonances. Actually, it can be more liberating than any imagined future if you are willing to identify its evasions, its distortions, its lies, and are willing to unleash its secrets.

- ☐ a) **Our present and future are hostages to our past, which has a lot to offer for those who are willing to analyse it.**
- ☐ b) **Understanding the past is a more fruitful exercise than imagining a future because of all the secrets the past carries within itself.**
- ☐ c) **There is a lot to learn from the past even though it is distorted by the present and is mostly unclear because of the lies it encompasses.**
- ☐ d) The past has an advantage over the future – it can be analysed for information.

Q19. DIRECTIONS *for questions 18 and 19:* The passage given below is followed by four summaries. Choose the option that best captures the essence of the para.

Companies can compete either by being first to develop and market new products or designs or by allowing industry innovators to bring out new products, then quickly copy successful characteristics. Since the early 1980s, the pace of change has accelerated, forcing managers to confront more carefully questions of timing in new product development and introduction. In many cases, first movers have benefited from capturing markets early on. In other cases, imitators have fared better. In a competitive environment, managers must choose between deepening their commitment to innovation or relinquishing their leadership position in favour of a strategy of imitation.

- ☐ a) **In a fast changing competitive scenario, managers must carefully choose between innovation or imitation because success and profitability accompany sometimes one, sometimes the other.**
- ☐ b) **Innovate and not imitate is the magic word for success. Managers must innovate to survive.**
- ☐ c) **At times, imitators fare better but on an overall basis, first movers have the upper hand and are likely to gain more market share.**
- ☐ d) **Imitators can never be market leaders; hence managers are always wary of this strategy despite there been evidence to the contrary.**

Q20. DIRECTIONS *for question 20:* The four sentences (labelled 1, 2, 3 and 4) below, when properly sequenced, would yield a coherent paragraph. Decide on the proper sequencing of the order of the sentences and key in the sequence of the four numbers as your answer.

- 1. They hold the unshakable conviction that individuals are extremely important, that every life matters.
- 2. They get excited over one smile; they are willing to feed one stomach, educate one mind, and treat one wound.
- 3. Sometimes they even transform cities and nations, and yes, the world.
- 4. I've noticed something about people who make a difference in the world.

Q21. DIRECTIONS *for question 21:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.

Sentence: It has to be the sea because the sea holds you up.

Paragraph: “My Body is a Cage”, goes the song by Arcade Fire: the problem is getting out of there without letting the door slam behind you, because the greatest escape of all, unfortunately, is death. There are day passes, of course: drugs, dissociative states, psychosis. There is alcohol, also meditation, the endorphin rush of exercise, the top of the mountain and the freewheel cycle downhill. _____(1)_____ Let us not forget poetry, the touch of someone you love; there is, when you think about it, love itself. But so effortful all of them, so exhausting, this battle through the ego to escape the ego, this pushing of the body to be free of the body, when all you really have to do to escape your sad sack of bones and flesh is take your clothes off, walk into the sea, and splash. _____(2)_____ It also slaps you and throws you about a bit, requiring your admiration, gratitude and respect. The sea is plentiful and cheap; you are, when you swim in it, connected to every other beach and rock you swam from over the years. _____(3)_____ It has to be the sea because there, chopped about by the advancing waves, is the horizon, and this does some secret, muscular thing to your gaze. _____(4)_____ After five minutes’ swimming towards it, your brain will open: simple as a window in the month of May.

☐ a) **Option 1**

☐ b) **Option 2**

☐ c) **Option 3**

☐ d) **Option 4**

Q22. DIRECTIONS *for question 22:* The four sentences (labelled 1, 2, 3 and 4) below, when properly sequenced, would yield a coherent paragraph. Decide on the proper sequencing of the order of the sentences and key in the sequence of the four numbers as your answer.

1. From now to the end of consciousness, we are stuck with the task of defending art.

2. None of us can ever retrieve that innocence before all theory when art knew no need to justify itself, when one did not ask of a work of art what it said because one knew (or thought one knew) what it did.

3. Indeed, we have an obligation to overthrow any means of defending and justifying art which becomes particularly obtuse or onerous or insensitive to contemporary needs and practice

4. We can only quarrel with one or another means of defence.

Q23. DIRECTIONS *for question 23:* Four alternative summaries are given below the text. Choose the option that best captures the essence of the text.

It is well known that the composer Beethoven represented calls of the cuckoo, quail and nightingale in the 'Pastoral' symphony. Beethoven, in fact, may have been such a fan of birdsongs that it is thought that he plagiarised a couple of motifs from contemporary feathered composers. At a convention, Baptista played a tape which included a phrase in Western music: "Ba-ba-ba baaahm." A high-pitched version of the opening of Beethoven's Fifth Symphony rang out unmistakably but Baptista said the notes actually came from a white-breasted wood wren in Mexico. Baptista also played other suspicious phrases, which form the lilting opening to the rondo of Beethoven's 'Violin Concerto in D, Opus 61'. A birder noted in 1953 that a European blackbird, a relative of U.S. robins, had come up with the same theme. Almost 30 years later, another sharp listener reported the same blackbird song. Both he and Baptista noted that generations of blackbirds seem to have preserved that tune, so perhaps it dated back to a time when Beethoven himself heard and borrowed it.

- ☐ a) **Beethoven has been accused of plagiarising many tunes from songbirds.**
- ☐ b) **The song of the songbird and the music created by Beethoven sound astonishingly similar and represent one of the many convergences Baptista has found between music and birdsongs.**
- ☐ c) **Just as Shakespeare borrowed from well-known stories, so Beethoven drew on the natural sounds and rhythms around him.**
- ☐ d) Beethoven, a composer, may have been inspired by birdsongs, as observed by Baptista and others.

Q24. DIRECTIONS *for question 24:* The four sentences (labelled 1, 2, 3 and 4) below, when properly sequenced, would yield a coherent paragraph. Decide on the proper sequencing of the order of the sentences and key in the sequence of the four numbers as your answer.

1. In the digital era of the 1970s, laser and inkjet printers arrived.

2. Since then, movable type has given way to other processes, such as lithography which was invented in 1796 and screen printing which was developed in 1910.

3. Printing has come a long way since Johannes Gutenberg perfected the commercial use of the printing press around 1439.

4. Then 3D printers emerged to make solid objects by building up layers of material.